
BAR: Budgeted Actor Refinement Decouples Bootstrap Exposure from Policy Reach

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Abstract

TD3+BC uses one actor branch for both Bellman targets and deployment, tying target-action movement to policy reach. Budgeted actor refinement (BAR) instead routes the first actor of a persistent K -actor chain into TD targets and deploys the last. We use *bootstrap exposure* operationally for target-action displacement, not critic error. A finite-step bound relates target-value perturbation to first-hop displacement, while an ideal Wasserstein-2 proximal analysis describes the local geometry. Across nine D4RL tasks, fourteen budgets, and four seeds, BAR’s bundled depth intervention raises the equally weighted $T \geq 4$ mean from 25.38 at $K = 1$ to 63.97 at $K = 4$ and cuts score- < 20 cells from 34/63 to 8/63; both disjoint seed pairs reproduce the full ordering. Its descriptive grid maximum is 82.06, versus 74.79 for TD3+BC. A 270-checkpoint audit finds lower P3 target-action displacement in 88/90 matched cells without typical final-policy contraction. Crucially, a two-actor control without sequential re-centering scores 59.22 versus 57.67 for a contemporaneous P3 rerun and has the same 9/45 seed-mean collapse count. On this grid, the results support target/deployment separation and the bundled depth-indexed procedure as effective finite-step stability controls, while leaving the chain’s incremental return benefit beyond separation as the decisive open ablation.

1 INTRODUCTION

Offline reinforcement learning (RL) seeks a policy from a fixed dataset without additional environment interaction. Its central difficulty is not merely to select a high-value action, but to determine how far policy improvement can proceed before the learned critic becomes unreliable. TD3+BC regularizes a deterministic actor toward dataset actions while using a Polyak copy of that actor in bootstrapped critic targets (Fujimoto and Gu, 2021). The same policy branch therefore determines both the action shift entering critic training and the reach of the evaluated policy. Aggressive improvement can move target actions into regions where critic error is amplified, whereas conservative improvement can leave useful policy improvement unrealized.

We study this coupling through *budgeted actor refinement* (BAR), a re-centered actor architecture for TD3+BC. BAR represents a nominal improvement budget with K sequential actor updates. The first actor is anchored to dataset actions, and each later actor is anchored to the stop-gradient output of its predecessor. A target copy of the first refined actor supplies the next actions in TD targets, while the final actor is used for evaluation. The architecture thus assigns critic-target construction and deployment to distinct policy branches without changing the critic update rule.

Let D_{critic} denote the squared displacement of the critic-coupled first-hop branch and let D_{final} denote that of the evaluated final policy. A one-hop actor identifies these quantities; BAR assigns them to μ_1 and μ_K , respectively. Throughout, *bootstrap exposure* denotes this target-action-displacement proxy, not distance to dataset support, critic error, or Bellman-target error. This separation can increase final-policy reach at comparable first-hop movement or contract both movements near an unstable training outcome. It also introduces a countervailing mismatch because the critic target follows the μ_1 branch while evaluation follows μ_K . The method therefore exposes a design axis between target-policy displacement and de-

ployment reach rather than guaranteeing that either quantity is independently optimal.

For deterministic actors, the squared TD3+BC anchor is a Wasserstein-2 movement cost in the Dirac-policy limit. With a fixed critic and small realized steps, ideal explicit and implicit updates discretize the same flow to first order. The large-budget tail instead probes the coupled finite-step regime, where projection, actor fitting, critic drift, saturation, and optimization history determine the end-to-end outcome.

The complete proximal phase diagram yields high-budget means $25.38 \rightarrow 41.20 \rightarrow 51.92 \rightarrow 63.97$ for $K = 1, 2, 3, 4$ and low-return counts $34 \rightarrow 22 \rightarrow 16 \rightarrow 8$. Both disjoint seed pairs reproduce the ordering; eight rescues account for 54.7% of the $K = 3 \rightarrow 4$ gain. A four-seed projected-linearized sweep reproduces the $K = 2 \rightarrow 3$ direction. Direct diagnostics show target-action contraction without typical final-policy contraction at moderate depth, while failed runs exhibit extreme TD-error tails, larger movement, and gain reversal under a common critic. A broader P3 control obtains the same aggregate collapse count with only directly data-anchored target and deployment actors, establishing policy separation as a sufficient candidate mechanism on this grid and making the chain’s incremental contribution an explicit experimental question.

Our contributions are as follows.

- We formulate BAR for TD3+BC as a re-centered actor composition that assigns target-policy displacement and final-policy reach to distinct branches.
- We give a Wasserstein proximal interpretation, derive an inexact finite-step margin with an explicit critic-error term, bound first-hop TD-target perturbation, and separate these statements from the fixed-critic small-step flow limit.
- Four-seed proximal sweeps through $K = 4$, a four-seed linearized sweep through $K = 3$, direct target-action measurements, and targeted controls show that the bundled depth intervention broadens the large-budget stability envelope; the largest proximal gain is driven primarily by lower-tail rescue, while a P3 two-actor control supports policy separation as a sufficient candidate mechanism.

2 RELATED WORK

Behavior-regularized offline actor-critic learning. Offline actor-critic methods must control value extrapolation when a learned policy selects actions

outside the fixed dataset. Prior work studies this failure through constrained policy improvement, conservative value learning, and restrictions on actions used in Bellman backups (Fujimoto et al., 2019; Kumar et al., 2019). TD3+BC provides a particularly minimal control: a deterministic actor maximizes a normalized critic while a squared cloning term anchors its actions to the dataset (Fujimoto and Gu, 2021). We keep this critic update and actor family fixed and study which refined actor branch supplies the TD target. One-step offline RL instead limits repeated policy optimization against an off-policy value estimate (Brandfonbrener et al., 2021). BAR retains iterative actor-critic training but distributes a nominal improvement coefficient across a routed actor chain.

Policy geometry and proximal refinement.

Trust-region policy optimization is commonly derived from information geometry (Amari and Nagaoka, 2000; Schulman et al., 2015). Optimal-transport geometry gives a different notion of movement: policy optimization can be expressed as a Wasserstein gradient flow, with related natural-gradient and trust-region constructions for continuous control (Zhang et al., 2018; Moskovitz et al., 2020; Terpin et al., 2022; Song et al., 2023). Wasserstein Policy Optimization projects Wasserstein policy flow into a parametric actor class using value-function action gradients (Pfau et al., 2025). For deterministic policies, the statewise Wasserstein-2 cost reduces to squared action displacement. We use that identity to interpret the TD3+BC anchor as a proximal movement cost, while separating the finite-step variational statement from its small-step flow limit.

Target networks and actor-critic coupling.

Bootstrapping error can feed critic extrapolation back into actor updates (Fujimoto et al., 2019; Kumar et al., 2019). Polyak averaging delays this feedback but ordinarily copies the same actor branch that will be deployed. MCEP instead pairs a conservative target policy for value learning with a mildly constrained evaluation policy excluded from Bellman backups (Xu et al., 2024). BAR shares this routing principle but uses a persistent, predecessor-recentered chain and varies depth under a fixed nominal budget; MCEP keeps both actors directly data-anchored with separately chosen strengths and is not a $K = 2$ instance of BAR. Our contribution concerns this depth-indexed routing operator rather than a new critic objective.

3 PRELIMINARIES

We consider a discounted Markov decision process $\langle \mathcal{S}, \mathcal{A}, P, r, \gamma \rangle$ with continuous actions and a fixed of-

fine dataset $\mathcal{D} = \{(s_i, a_i, r_i, s'_i)\}_{i=1}^N$. A clipped double critic is trained from targets

$$y_\rho = r + \gamma \min_j Q_j^-(s', \rho(s')), \quad (1)$$

where ρ is a target policy. When $\rho(s')$ leaves dataset support, critic error enters the regression target and can be amplified by later actor updates (Fujimoto et al., 2019; Kumar et al., 2019).

For a deterministic actor μ_θ , normalized TD3+BC minimizes

$$\mathcal{L}_{\text{actor}}(\theta) = -\alpha \mathbb{E}_s \bar{Q}(s, \mu_\theta(s)) + \mathbb{E}_{(s,a) \sim \mathcal{D}} \frac{\|\mu_\theta(s) - a\|_2^2}{d}, \quad (2)$$

where $\bar{Q} = \hat{Q}/C$ and $C > 0$ is a stop-gradient critic scale (Fujimoto and Gu, 2021). Ordinarily, a Polyak copy of this actor enters Eq. (1), while the online actor is deployed. Target-policy movement and deployment reach therefore follow the same branch.

A metric proximal update with step $h > 0$ takes the form

$$\pi^+ \in \arg \min_{\pi} \left\{ \hat{\mathcal{E}}(\pi) + \frac{1}{2h} d_{\mathcal{M}}^2(\pi, \pi^{\text{ref}}) \right\}. \quad (3)$$

This minimizing-movement construction is associated with metric gradient flows (Ambrosio et al., 2005; Otto, 2001). For deterministic state-conditioned policies,

$$W_2^2(\delta_{\mu(s)}, \delta_{\nu(s)}) = \|\mu(s) - \nu(s)\|_2^2. \quad (4)$$

Thus the TD3+BC anchor is a Wasserstein-2 movement cost up to fixed scaling. We distinguish its finite-step variational meaning from the small-step, fixed-critic gradient-flow limit and from large-budget coupled training.

4 BUDGETED ACTOR REFINEMENT

BAR maintains K independently initialized, persistent deterministic actors with separate optimizer states. At each delayed update they take one Adam step in chain order on the same minibatch; Figure 1 gives the exact live ordering. The online critic is fixed within the actor chain; we denote its first Q-function by $\hat{Q}$. We first derive the proximal operator motivating these updates and then state its live-algorithm boundary.

4.1 Offline actor updates as proximal improvement

Let $\rho_{\mathcal{D}}$ be the empirical state distribution and consider a deterministic actor $\mu : \mathcal{S} \rightarrow \mathcal{A} \subset \mathbb{R}^d$. At one training iteration, the critic parameters are held fixed while the

actor chain is updated. We write the current critic as $\hat{Q}_t$, use $\epsilon = 10^{-6}$, and define the implementation-level proximal scales by

$$\begin{aligned} C_{1,t}^{\text{P}} &= \text{sg} \left(\mathbb{E}_s |\hat{Q}_t(s, \mu_{1,t}^{\text{pre}}(s))| + \epsilon \right), \\ C_{k,t}^{\text{P}} &= \text{sg} \left(\mathbb{E}_s |\hat{Q}_t(s, \text{sg } \mu_{k-1,t}^+(s))| + \epsilon \right), \quad k \geq 2. \end{aligned} \quad (5)$$

The normalized critic is $\bar{Q}_{k,t}(s, a) = \hat{Q}_t(s, a)/C_{k,t}^{\text{P}}$. Here *pre* and the superscript $+$ denote actions before and after the current actor update, and *sg* denotes stop-gradient. We suppress t and the superscript when discussing the ideal operator. A fixed positive C only rescales gradient-flow time; recomputing C_k changes the finite-step schedule. Our finite-step statements condition on the realized scales.

For deterministic policies and a state distribution ρ , define the action-averaged Wasserstein metric

$$\mathbf{d}_{d,\rho}^2(\mu, \nu) := \mathbb{E}_{s \sim \rho} \left[\frac{1}{d} \|\mu(s) - \nu(s)\|_2^2 \right]. \quad (6)$$

It is d^{-1} times the state-averaged squared W_2 distance between state-conditioned Dirac policies. Strictly, it is a pseudometric on pointwise policy maps and a metric after identifying maps equal ρ -almost everywhere. We abbreviate $\mathbf{d}_d = \mathbf{d}_{d,\rho_{\mathcal{D}}}$ on the actor-training state distribution. The population counterpart of squared behavior cloning also admits a deterministic reference. Let P_B be the data-generating behavior distribution, let ρ_B be its state marginal, and define $b(s) = \mathbb{E}_{P_B}[a | s]$. Then

$$\mathbb{E}_{(s,a) \sim P_B} \frac{1}{d} \|\mu(s) - a\|_2^2 = \mathbf{d}_{d,\rho_B}^2(\mu, b) + \text{const}, \quad (7)$$

where the constant is the conditional action variance and does not depend on μ . The empirical actor loss is a Monte Carlo estimate of the left-hand side, so samplewise squared anchoring targets the conditional-mean behavior map at the population level.

Given a reference policy ν , step $h > 0$, and fixed normalized critic $\bar{Q}$, define the proximal refinement operator by

$$\mathcal{P}_h^{\bar{Q}}(\nu) \in \arg \min_{\mu \in \mathcal{M}} \left\{ -\mathbb{E}_s \bar{Q}(s, \mu(s)) + \frac{1}{2h} \mathbf{d}_d^2(\mu, \nu) \right\}. \quad (8)$$

At hop k , setting $\bar{Q} = \bar{Q}_k$ and multiplying by $2h$ gives the practical actor objective

$$\mathcal{L}_{h,k}(\mu | \nu) = -2h \mathbb{E}_s \bar{Q}_k(s, \mu(s)) + \mathbf{d}_d^2(\mu, \nu). \quad (9)$$

Proposition 1 (Behavior anchoring is Wasserstein proximal improvement). *For deterministic policies, the normalized anchored objective*

$$-\alpha \mathbb{E}_s \bar{Q}(s, \mu(s)) + \mathbf{d}_d^2(\mu, \nu)$$

One BAR training iteration. The K actors are persistent, independently initialized, and have separate Adam states. 1. Form the Bellman target with μ_1^- and update the online critic. 2. On a delayed step, update μ_1 once toward sampled dataset actions. 3. Polyak-update μ_1^- and the target critics with rate .005. 4. For $k = 2:K$, use the updated sg μ_{k-1}^+ as reference and update μ_k once on the same mini-batch, holding the online critic fixed. 5. Use μ_1^- in future Bellman targets and deploy μ_K.	
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Figure 1: Actual update and routing order. Each hop is one optimizer step, not an exact proximal solve.

is Eq. (8), up to multiplication by a positive constant, with $h = \alpha/2$. Equivalently, the raw-critic objective $-\lambda \mathbb{E}_s \hat{Q}(s, \mu(s)) + d_d^2(\mu, \nu)$ has $h = \lambda C/2$ when C is fixed. The population counterpart of squared dataset anchoring uses $\nu = b$ by Eq. (7).

The proposition identifies an update operator relative to the *estimated* critic; true-return improvement lies outside the scope of this certificate.

4.2 Re-centered composition and actor-critic dataflow

For population analysis, let $\mu_0 := b$ denote the conditional-mean behavior map and let $h_{1:K}$ be positive step sizes. The idealized BAR chain is

$$\mu_k \in \mathcal{P}_{h_k}^{\bar{Q}_k}(\text{sg } \mu_{k-1}), \quad k = 1, \dots, K. \quad (10)$$

The implementation approximates each subproblem with one Adam update. Its first actor μ_1 uses sampled dataset actions as its reference; later actors $\mu_2, \dots, \mu_K$ use the preceding online actor under the same current critic. Only the Polyak copy of μ_1 enters the TD target, while μ_K supplies the evaluated action. Because the later persistent actors are independently initialized, their anchoring gradient need not vanish even at $h = 0$ when predecessor and current actor differ. The ideal proximal map therefore motivates, but does not certify, the live one-Adam-step chain.

Our fixed-budget schedule uses the implementation parameter $T = \tau = \alpha/2$, sets $h_k = T/K$, and holds T fixed. Thus $K = 1$ recovers the TD3+BC coefficient $\alpha = 2T$; T is neither a compute budget nor a realized distance constraint. Increasing K jointly changes local coefficient, re-centering frequency, actor count and compute, normalization trajectory, and target/deployment dataflow. Each K is therefore a complete randomized procedure, and its seed average

estimates marginal end-to-end performance. Identical critic initialization across depths is a common-random-number variance-reduction design, not a validity condition.

On the next-state distribution $\rho'_{\mathcal{D}}$ used by the TD target, define the action-averaged population displacements

$$\begin{aligned} D_{\text{critic}} &:= d_{d, \rho'_{\mathcal{D}}}^2(\mu_1, \mu_0), \\ D_{\text{final}} &:= d_{d, \rho'_{\mathcal{D}}}^2(\mu_K, \mu_0). \end{aligned} \quad (11)$$

A one-step actor has $D_{\text{critic}} = D_{\text{final}}$; BAR assigns the two population quantities to distinct policy branches. On fixed validation batches, we measure the sample-anchored online branches and the deterministic next-state Polyak target branch:

$$\begin{aligned} \hat{D}_{\text{critic}} &= \mathbb{E}_i \|\mu_1(s_i) - a_i\|_2^2/d, \\ \hat{D}_{\text{final}} &= \mathbb{E}_i \|\mu_K(s_i) - a_i\|_2^2/d, \\ \hat{D}_{\text{target}} &= \mathbb{E}_i \|\mu_1^-(s'_i) - a_{i+1}\|_2^2/d. \end{aligned} \quad (12)$$

The direct audit retains only bootstrapped transition pairs; a smoothed version uses common TD3 target-noise draws. Sample anchoring adds a common conditional-action-variance floor in population differences. These are movement measurements, not critic-reliability estimates.

4.3 Finite-step statements

For a re-centered hop where μ_{k-1} is feasible, suppose the realized actor satisfies $\mathcal{L}_{h_k, k}(\mu_k \mid \mu_{k-1}) \leq \mathcal{L}_{h_k, k}(\mu_{k-1} \mid \mu_{k-1}) + \varepsilon_k$. Rearranging the practical objective gives the conditional estimated-critic margin

$$\begin{aligned} m_k^{\bar{Q}} &:= \frac{2h_k}{C_k} \mathbb{E}_s [\hat{Q}(s, \mu_k(s)) - \hat{Q}(s, \mu_{k-1}(s))] \\ &\quad - d_d^2(\mu_k, \mu_{k-1}) \geq -\varepsilon_k. \end{aligned} \quad (13)$$

For $\hat{Q} = Q^\dagger + e$, this transfers to a trusted objective as

$$m_k^{Q^\dagger} = m_k^{\bar{Q}} - \frac{2h_k}{C_k} \mathbb{E}_s [e(s, \mu_k(s)) - e(s, \mu_{k-1}(s))]. \quad (14)$$

The first sample-anchored hop lacks a feasible deterministic dataset-action comparator, and one Adam step supplies no bound on the unmeasured ε_k . These statements therefore diagnose a realized learned objective; they guarantee neither exact proximal solution nor monotone return. The Supplement gives the derivation and bounded-error corollary.

Proposition 2 (First-hop TD-target perturbation bound). *For clarity, consider a noiseless TD target using μ_1 , $y_{\mu_1} = r + \gamma \min_j Q_j^-(s', \mu_1(s'))$, and suppose*

each target critic $Q_j^-(s', \cdot)$ is $L(s')$ -Lipschitz. For any reference μ_0 ,

$$\|y_{\mu_1} - y_{\mu_0}\|_{L_2(\mathcal{D})} \leq \gamma \left(\mathbb{E}_{s'} L(s')^2 \cdot \|\mu_1(s') - \mu_0(s')\|_2^2 \right)^{1/2} \quad (15)$$

If $L(s') \leq \bar{L}$, the right-hand side is at most $\gamma \bar{L} \sqrt{d D_{\text{critic}}}$. Conditional on μ_1 , later actors do not enter this one-step target perturbation.

Proposition 2 bounds a one-step value-target difference, not the stability of the learned critic. Its architectural content is that later refinement can alter D_{final} without directly entering this bound. With a Polyak target network, μ_1 is replaced by its target copy. A same-noise comparison under TD3 target smoothing obeys the same bound because action clipping is non-expansive. The current-state $\hat{D}_{\text{critic}}$ omits Polyak lag and target noise, whereas $\hat{D}_{\text{target}}$ measures the Polyak actor directly and has a smoothed counterpart. Neither displacement measures the action Lipschitz factor or the resulting change in critic value.

The separation introduces a complementary evaluation-policy mismatch: critic targets follow the μ_1 branch, whereas evaluation follows μ_K . Increasing D_{final} at fixed D_{critic} can therefore make the critic less representative of the evaluated policy. We interpret BAR as exposing a measurable target/deployment trade-off; its empirical return captures the net effect of target movement, evaluation mismatch, actor compute, and the coupled optimization trajectory.

4.4 Scope of the local idealization

With a fixed critic, inactive projection, small realized displacement, and an exact statewise solve, the action-metric linearized step is βg and the ideal proximal step is $\beta g + \beta^2 Hg + O(\beta^3)$ for $\beta = dh/C$ and locally Lipschitz Hessian. Thus both are first-order local schemes; a generic consistent map has fixed- K flow bias of order T^2/K . The full expansion is in the Supplement. It is not an explanation of the headline regime: persistent later actors need not approach the identity as $h \rightarrow 0$, and large- T training changes projection, fitting, critic, scale, and optimizer history. For the learned ReLU critic, the fixed-state action gradient is piecewise constant: the two ideal maps coincide inside one activation region, while a boundary crossing exits the smooth analysis. On the locked Hopper/Walker twelve-run residence audit (6144 anchors; independent verifier pass), empirical full-horizon residence is 0.954, 0.911, 0.842, and 0.748 at $T \in \{.025, .05, .1, .2\}$. These rates delimit the local affine scope; they are not return

effects, global proximal certificates, or a learned $1/K$ slope.

5 EXPERIMENTS

We evaluate BAR on the nine D4RL locomotion tasks formed by HalfCheetah, Hopper, and Walker2d with medium, medium-replay, and expert datasets (Fu et al., 2020). These tasks directly exercise deterministic continuous control and provide distinct behavior-data regimes for a target-routing stress test. Base implementations follow CORL configurations (Tarasov et al., 2022); scores are D4RL normalized returns. Additional per-task results and diagnostic definitions are in the Supplementary Material.

Fixed-budget protocol and estimand. We hold a total nominal budget T fixed, split it across K actor hops as $h = T/K$, and sweep

$$T \in \{0.05, 0.1, 0.2, 0.4, 0.7, 1.5, 2.5, 4, 7, 10, 12, 14, 17, 20\}.$$

over all nine tasks. Both grids use seeds 0–3, for $9 \times 14 \times 4 = 504$ runs per depth. Proximal seeds 0–1 and 2–3 were executed on separate hosts and form disjoint replication pairs. The grid is a high-resolution stability phase diagram rather than a final benchmark alone. For $K = 1$, Proposition 1 maps T to the normalized TD3+BC coefficient $\alpha = 2T$.

We compare TD3+BC ($K = 1$), proximal BAR with $K \in \{2, 3, 4\}$ (BAR-Prox), and an action-metric-matched projected linearized target with $K \in \{2, 3\}$ (BAR-Lin):

$$a_k^{\text{tar}} = \Pi_{\mathcal{A}} \left[\nu_k + d \frac{T}{K} \frac{\nabla_a \hat{Q}(s, \nu_k)}{C_{k,t}^L} \right], \quad (16)$$

$$\nu_1 = a, \quad \nu_k = \text{sg } \mu_{k-1,t}^+(s) \quad (k \geq 2).$$

At each hop, BAR-Lin detaches the reference, recomputes the $\hat{Q}$ gradient and scale, clips the target to the action box, and takes one mean-squared-error optimizer step with the corresponding persistent actor. Its first-hop scale is

$$C_{1,t}^L = \text{sg} \left(\mathbb{E}_{(s,a) \sim \mathcal{D}} |\hat{Q}_t(s, a)| + \epsilon \right),$$

whereas $C_{k,t}^L = C_{k,t}^P$ for $k \geq 2$. Thus the factor d matches the action metric, but the first-hop normalization point differs from BAR-Prox; the comparison does not match the complete scale schedule. It uses the same first-hop target and final-hop evaluation routing. All conditions match datasets, critic updates, and the 10^6 -update horizon; every headline score is the ten-episode D4RL normalized mean at the final checkpoint. Intermediate evaluation cadence varies by run

Table 1: Aggregate fixed-budget results. Every entry uses four seeds. Best (T) is the descriptive maximum over the shared 14-value grid; High denotes $T \geq 4$. Collapse entries are seed-mean cells / raw seeded runs.

Method	$T = 1.5$	Best (T)	High	Collapse
TD3+BC	74.79	74.79 (1.5)	25.38	34 / 142
BAR-P2	78.05	80.66 (2.5)	41.20	22 / 111
BAR-P3	78.95	81.41 (2.5)	51.92	16 / 84
BAR-P4	77.70	82.06 (4)	63.97	8 / 54
BAR-L2	75.13	78.43 (2.5)	38.43	25 / 112
BAR-L3	77.74	78.20 (2.5)	46.44	21 / 95

family and does not enter this estimand. A K -hop condition performs K actor optimizer steps per delayed update, so depth comparisons are between complete training procedures. The end-to-end intervention changes local coefficient, re-centering frequency, actor compute, normalization trajectory, routing, and optimization history.

We test whether this intervention reduces large-budget low-return outcomes, persists under the linearized realization, separates first-hop exposure from final-policy reach, and exposes critic-validity signatures of failure.

5.1 Additional actors broaden the stability envelope

Figure 2 and Table 1 summarize the sweep. The high-budget estimand averages seeds and the seven $T \geq 4$ budgets within each task, then weights tasks equally. A seed-mean environment-budget cell is descriptively collapsed below score 20. Across proximal $K = 1, 2, 3, 4$, the mean rises $25.38 \rightarrow 41.20 \rightarrow 51.92 \rightarrow 63.97$ while collapsed cells fall $34 \rightarrow 22 \rightarrow 16 \rightarrow 8$ of 63. $K = 4$ gains 12.06 over $K = 3$ with task-resampling interval $[3.84, 21.11]$ and seven of nine positive task means, yet its cellwise median is only $+0.34$: eight rescues, no reversals, contribute 54.7% of the gain. The $T \geq 10$ tail remains strictly ordered, $20.27 \rightarrow 31.01 \rightarrow 42.23 \rightarrow 56.79$. BAR-Lin $K = 3$ raises the four-seed high-budget mean from 38.43 to 46.44 and reduces collapsed cells from 25 to 21; its $+1.37$ cellwise median indicates a broader shift. The largest proximal gain therefore acts chiefly through lower-tail rescue rather than a uniform shift.

The high-budget aggregate is a robustness estimand, not a deployment rule. At TD3+BC’s grid-best shared budget, $T = 1.5$, BAR-P3 and P4 score 78.95 and 77.70 versus 74.79. Allowing one globally shared grid value per method, BAR-P4 reaches 82.06 at $T = 4$. In a post-hoc tuning-transfer sensitivity that selects each method’s T on two dynamics families and

tests on the third, P4 averages 78.44 versus 74.79 for TD3+BC ($+3.65$; 7/9 task differences positive). Family gains vary, so these summaries demonstrate a practical regime but do not provide an offline selection rule.

Task dependence and replication. Responses range from smooth to sharply nonmonotone, so there is no universal critic-valid radius. Nominal T is the strongest absolute within-environment rank predictor of return in eight of nine environments; realized movement is a mechanism readout, not a substitute for the intervention. P4–P3 is positive within all three dynamics families ($+0.79, +13.04, +22.33$), and every leave-one-family-out mean remains positive ($+6.92$ to $+17.69$). Both host-separated seed pairs reproduce the high-budget ordering: 25.97, 42.02, 52.78, 63.61 and 24.79, 40.38, 51.05, 64.33. These are disjoint-seed replications of one implementation.

The linearized sweep shows that the gain is not unique to the proximal loss. At high budgets, BAR-Lin with $K = 3$ gains 8.01 points over $K = 2$ (95% task-resampling interval $[3.56, 13.53]$; eight of nine positive task means) and reduces collapse cells from 25 to 21. Both disjoint seed pairs reproduce the direction, with gains of 7.23 and 8.80. BAR-Prox remains stronger overall, so these are comparisons of complete training procedures rather than isolated integrator effects.

5.2 Large-budget behavior is nonlocal and critic dependent

At $T = 20$, the projected BAR-Lin $K = 2$ target is clipped at an action bound in 94–95% of states, and the effective action-step coefficient h_{eff} is only 14% and 9% of the nominal dT/K coefficient in the two hops. These are medians over 18 environment–seed runs per hop. The practical operation is projected target generation followed by amortized neural regression under a moving critic. The sweep measures the resulting end-to-end stability envelope; local truncation error lies outside this estimand.

Figure 3 shows the common failure signature. Environment return supplies the external label: collapsed runs score below 20. These runs have orders-of-magnitude larger TD-error tails, larger critic magnitudes, and larger realized movements. The controlled common critic is the frozen final TD3+BC $T = 1$ critic matched by environment and seed. A run-total gain sums the mean Q change across actor hops on the shared state batch. For collapsed BAR-Lin runs, its median under the jointly trained critic is 3.33×10^{11} ; under the common critic it is -1.42 , compared with $+2.11$ for stable runs. For BAR-Prox, the common-critic gain changes from $+2.51$ in stable runs to -1.06 in collapsed runs. This sign reversal diagnoses cross-

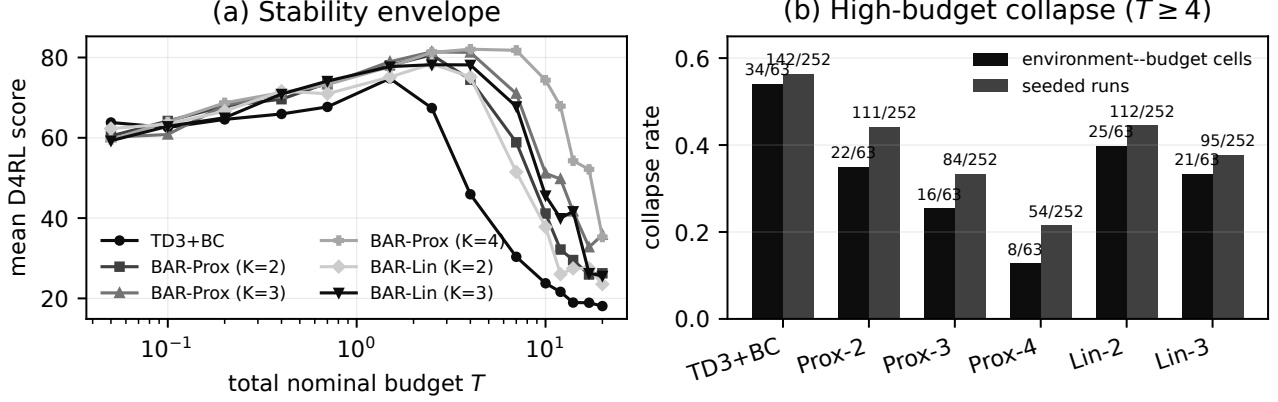


Figure 2: Fixed-budget sweep over nine locomotion tasks. (a) Mean score over tasks. (b) Collapse rates for $T \geq 4$, reported both by environment-budget cell and by seeded run. All curves and cell means use four seeds. Both realizations broaden the stable high-budget region with depth.

critic disagreement consistent with Eq. (14). A separate four-environment seed-2/3 sensitivity audit, using a fixed TD3+BC $T = .05$ reference critic, preserves the split: all 45 stable gains are positive and all 35 collapsed gains are negative (Supplement).

A simulator-reference audit provides targeted evidence of critic miscalibration in the screened checkpoints: the median signed error $\hat{Q}_1(s, a_K) - Q_{MC}^{\rho_1}(s, a_K)$ is $+1.70 \times 10^{12}$ in five collapsed checkpoints and -41.4 in 55 stable ones, using the median of checkpoint-level state medians. Episode-cluster intervals exclude zero for critic error at 58 of 60 checkpoints, and restoring the sampled episode clock preserves the error sign at all 60. Its action-ranking subset is sparse, so we use it only as targeted checkpoint-level evidence.

5.3 Target-action displacement and final-policy reach

At the same nominal budget, BAR-Prox realizes a smaller current-state first-hop displacement proxy than BAR-Lin once $T \geq 7$ (the Supplementary Material). At $T = 20$, their median $\hat{D}_{critic}$ values are 0.195 and 0.323. Across 90 matched environment-budget-seed cells at $T \in \{4, 7, 10, 14, 20\}$, exactly one of BAR-Prox and BAR-Lin collapses in 11 cases. In all 11, the stable realization has both smaller $\hat{D}_{critic}$ and smaller $\hat{D}_{final}$. These matched cells associate smaller current-state movement with stable trajectories; they do not isolate a causal TD-target route.

We measure Eq. (12) on all 270 audited method checkpoints. The current-state first-branch proxy closely tracks the deterministic target-action metric ($r = .9999$). BAR-Prox $K = 3$ has lower sample-anchored target-action displacement than TD3+BC in 88 of 90

matched cells (median ratio .658), while the current-state $\hat{D}_{final}$ proxy is lower in only 36 of 90 (median ratio 1.057). A few large final-policy contractions make the arithmetic mean difference negative despite this typical-cell expansion. Thus audited three-hop BAR exhibits the intended asymmetric movement response rather than merely shrinking both branches. The P4 target-branch archive contains 180 checkpoints across seeds 0–3. Comparisons use the 90 seed-0/1 cells matched to TD3+BC and P3: P4 has lower deterministic next-state target-action displacement in 89/90 cells versus TD3+BC (median ratio .568) and 85/90 versus P3 (.950). The archive omits P4 final-actor geometry; that comparison is limited to a separate 24-cell frontier panel. Added depth therefore contracts the measured target-action proxy without establishing full-grid exposure-reach decoupling or universal final-policy contraction. The P3/P2 direction also holds in 36/40 cells of a four-environment seed-2/3 sensitivity (median ratio .905). Yet exposure alone does not determine return: P3 loses by more than one point in 17 of the original 84 lower-exposure cells (Supplement).

A targeted Hopper-medium audit exhibits both forms of decoupling. Over $T = 4$ –10, $K = 3$ extends final-policy reach at nearly unchanged first-hop movement; near the $T = 12, 14$ transition it contracts both movements and avoids the low-return outcome. At $T = 20$, both depths saturate and fail (Supplement).

5.4 Targeted component controls

To test whether sequential re-centering is required, we use a broader two-seed control whose target and deployed actors are directly data-anchored with coefficients $T/3$ and T . Table 2 compares it only with a contemporaneous BAR-P3 rerun. Control minus BAR

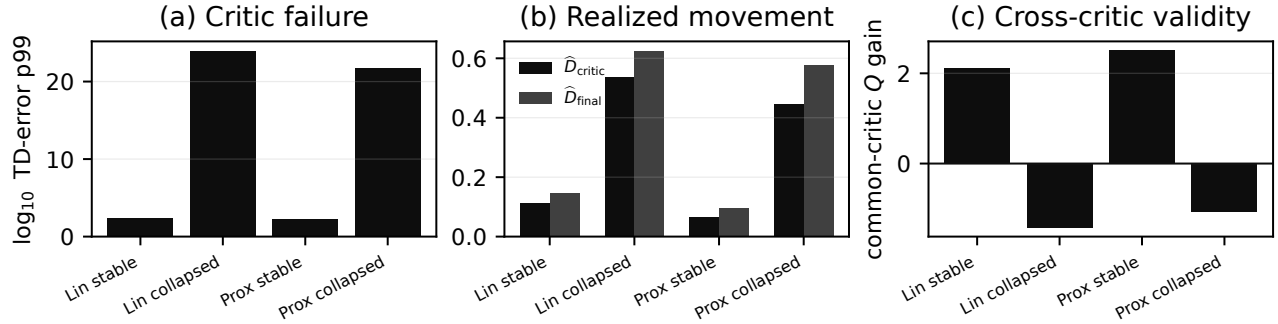


Figure 3: Median diagnostics over the matched $T \in \{4, 7, 10, 14, 20\}$ audit. Proximal and linearized realizations share a diagnostic pattern consistent with critic miscalibration: critic errors and realized movements increase sharply, and run-total gains under a common critic become negative. Environment return labels failure; the common critic diagnoses cross-critic disagreement.

Table 2: P3 policy-separation control over 90 matched task-budget-seed pairs. Collapse is the descriptive score-20 count before / after seed averaging.

Condition	Mean	Raw < 20	Cell < 20
BAR-P3 rerun	57.67	23/90	9/45
Two-actor control	59.22	21/90	9/45

is +1.56 with task-resampling interval $[-1.52, 5.44]$, while its paired median is $-.42$ and BAR wins 52/90 pairs; only 2/9 task means favor the control. Thus policy separation attains the observed P3 aggregate regime without sequential re-centering. This is a bundled, non-compute-matched comparison that also changes anchoring and critic trajectory, not an equivalence result or a re-centering-only effect. No corresponding P4 control has yet tested the source of P4’s additional gain.

Smaller post-hoc controls on eight frontier runs find similar scores for compute-matched sequential and fixed-reference variants (71.4 and 70.0), while direct-final refinement scores 59.4; slower target averaging alone does not rescue $K = 1$. Actor-detached route shadows do not meet their pre-specified common-continuation rule and estimate no return effect, so their calibration results remain in the Supplement.

5.5 Discussion and limitations

Both realizations broaden the high-budget stability envelope, with BAR-Prox stronger. Within the bundled K intervention, P3 exhibits asymmetric target and final displacement, while P4 adds target-branch contraction and lower-tail rescue. The P3 control identifies policy separation as a sufficient candidate mechanism but leaves the chain’s incremental P4 contribu-

tion unresolved. A projection-free frozen-critic audit supports local explicit-implicit agreement; the live result belongs to a coupled finite-step procedure in which depth changes local step, re-centering, actor compute, normalization, routing, and optimization history.

Both headline score grids use four seeds, and the proximal grid forms two host-separated pairs. The full P3 movement and exposure audit uses seeds 0–1; a restricted four-environment seed-2/3 exposure audit supplies a sensitivity check, while the P4 target dump uses four seeds. For proximal seeds 2–3, provenance consists of the archived launch configuration and score matrices; exact per-run manifests were not retained, and post-hoc P4 manifests do not reconstruct them. Only a 24-cell panel measures matched P4 final displacement. The tasks span three dynamics families, collapse is descriptive, and controls are post hoc. The 90-pair policy-separation control uses seeds 0–1, is not compute matched, and includes ten Walker2d-expert cells completed through a documented CPU recovery chain (eight resumed and two CPU-only). We provide no offline rule for choosing T or K . BAR makes K actor optimizer calls per delayed update and stores K persistent actors, so actor work and persistent actor state scale as $O(K)$. On one H200/JAX 0.10.2 stack, compiled actor-phase time rises from 1.435 to 1.887 ms and scoped backend high-water memory from 9.68 to 18.05 MB between $K = 1$ and $K = 4$ ($1.316\times$, $1.863\times$); these one-stack measurements are not end-to-end scaling. The ideal fixed-critic analysis motivates but does not certify the live chain.

6 CONCLUSION

BAR demonstrates a substantially broader offline actor-critic stability envelope for a procedure that routes target and deployment movement through dis-

tinct branches. Across four proximal seeds, the $T \geq 4$ mean rises from 25.38 at $K = 1$ to 63.97 at $K = 4$ as low-return cells fall from 34/63 to 8/63; both disjoint seed pairs reproduce the ordering, and BAR-P4 also attains the strongest descriptive grid maximum. The four-seed linearized grid reproduces the $K = 2 - K = 3$ direction. Direct audits identify asymmetric target and final displacement at moderate depth and added-depth first-hop contraction, with lower-tail rescue driving most of the latter gain. Within BAR, K is therefore an effective finite-step procedure control, not merely a discretization-accuracy knob. The two-actor result identifies policy separation as the simpler sufficient mechanism hypothesis; a matched P4 control must resolve the sequential chain’s incremental value.

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Checklist

- For all models and algorithms presented, check if you include:
 - A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes/No/Not Applicable]
Yes. Sections 3–4 define the setting, actor chain, losses, and assumptions.
 - An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes/No/Not Applicable]
Yes. Section 4 states that BAR uses K persistent actors and K actor optimizer calls per delayed update; Eq. (13), Proposition 2, and the local-idealization subsection state the theoretical properties and scope. The supplement reports a verified one-stack time and memory profile.
 - (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes/No/Not Applicable]
Partial. The anonymized release contains the BAR launch, evaluation, diagnostic, and

verification code plus the compact policy-separation audit. The control-run configs identify local training-source files by SHA-256, but did not record a clean Git revision.

2. For any theoretical claim, check if you include:

- (a) Statements of the full set of assumptions of all theoretical results. [Yes/No/Not Applicable]

Yes. Each proposition states its assumptions, and the local-flow assumptions are separated from the coupled large-budget regime.

- (b) Complete proofs of all theoretical results. [Yes/No/Not Applicable]

Yes. Complete proofs are provided in the supplementary material.

- (c) Clear explanations of any assumptions. [Yes/No/Not Applicable]

Yes. Sections 4 and 5 explain the learned-critic, fixed- K , small-step, and implementation restrictions.

3. For all figures and tables that present empirical results, check if you include:

- (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Yes/No/Not Applicable]

Yes. Released CSVs, figure-generation code, a numerical claim verifier, and build instructions reproduce every main figure and aggregate table.

- (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes/No/Not Applicable]

No. All recoverable settings are listed, but exact per-run BAR-P4 configurations and manifests for proximal seeds 2–3 are unavailable; the policy-separation configs also omit a source revision and environment lock.

- (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes/No/Not Applicable]

Yes. Captions and text define the equally task-and-budget-weighted high-budget estimand, seed-mean cells, raw seeded runs, collapse thresholds, paired ratios, and task-resampling intervals. Both headline grids use four seeds; proximal seeds form two host-separated pairs. Mechanism audits use the explicitly reported two- or four-seed subsets.

- (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes/No/Not Applicable]

No. The released artifact fully records the H200 used for the actor-cost profile and the policy-separation CPU recovery chain, but does not contain a complete, independently verified inventory of all training hardware.

4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:

- (a) Citations of the creator If your work uses existing assets. [Yes/No/Not Applicable]

Yes. D4RL, TD3+BC, and the CORL implementation basis are cited.

- (b) The license information of the assets, if applicable. [Yes/No/Not Applicable]

Yes. The code release identifies its license; upstream dependencies retain their own licenses.

- (c) New assets either in the supplemental material or as a URL, if applicable. [Yes/No/Not Applicable]

Yes. An anonymized code and compact-results artifact accompanies the submission.

- (d) Information about consent from data providers/curators. [Yes/No/Not Applicable]

Not Applicable. We use public simulated-control benchmark datasets and collect no human data.

- (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Yes/No/Not Applicable]

Not Applicable. The assets contain simulated locomotion trajectories and no personal or sensitive content.

5. If you used crowdsourcing or conducted research with human subjects, check if you include:

- (a) The full text of instructions given to participants and screenshots. [Yes/No/Not Applicable]

Not Applicable.

- (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Yes/No/Not Applicable]

Not Applicable.

- (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Yes/No/Not Applicable]

Not Applicable.